

A Primer on the Implemented Korea Spatial GE Model

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1 Purpose

This note documents the quantitative model implemented in the `korea_growth` package. It is a background primer rather than a full theory paper: the goal is to state the objects, equilibrium conditions, and numerical solution method that are actually used by the code.

The implementation solves a sequence of static spatial equilibria. The only dynamic state is lagged population, so period t equilibrium takes L_{t-1} as given and returns $(w_t, r_t, P_t, E_t, L_t, \bar{\tau}_t, \bar{\pi}_t)$.

2 Environment

Time is discrete, $t = 1, \dots, T$. Regions are indexed by $o, d \in \mathcal{N}$ and sectors by $j, k \in \mathcal{J}$. Let $N = |\mathcal{N}|$ and $J = |\mathcal{J}|$.

The exogenous inputs in period t are:

- regional amenities $\bar{V}_{d,t}$,
- domestic trade costs $\tau_{odj,t} \geq 1$ and import costs $\tilde{\tau}_{oj,t} \geq 1$,
- migration costs $\delta_{od,t} \geq 1$,
- sectoral productivities $A_{oj,t}$,
- land shares $\beta_{oj,t}$ and value-added shares $\gamma_{oj,t}$,
- input-output shares $\gamma_{ojk,t}^{IO}$,
- fixed costs $F_{oj,t}$, export fixed costs $\tilde{F}_{oj,t}$, and adoption fixed costs $\check{F}_{oj,t}$,
- subsidy rates $s_{oj,t}$,
- foreign demand shifters $\tilde{D}_{j,t}$ and foreign prices $\tilde{p}_{j,t}$,
- land or structures supply $H_{o,t}$,
- the mass of potential firms $M_{j,t}$.

For the agriculture sector there is an alternative mechanized technology with $(\check{\beta}_{oj,t}, \check{\gamma}_{oj,t}, \check{\gamma}_{ojk,t}^{IO})$ and efficiency gain $\xi > 1$.

The structural parameters are:

$$\sigma > 1, \quad \theta > 0, \quad \kappa > 1, \quad \rho_j \in (0, 1), \quad \iota \in (-1, 0), \quad \nu > 0, \quad \eta \in \mathbb{R},$$

together with expenditure parameters α_j and v_j satisfying $\sum_j \alpha_j = 1$ and $\sum_j v_j = 0$.

3 Households

3.1 Disposable income and indirect utility

Per-capita disposable income in region d is

$$y_{d,t}^{pc} = (1 - \bar{\tau}_t + \bar{\pi}_t)w_{d,t}, \quad (1)$$

where $\bar{\tau}_t$ is the proportional tax needed to finance subsidies and infrastructure and $\bar{\pi}_t$ is aggregate profit income rebated in wage units.

Given sectoral price indices $P_{dj,t}$, define the composite price index

$$\mathcal{P}_{d,t} = \prod_{j \in \mathcal{J}} P_{dj,t}^{\alpha_j}. \quad (2)$$

The implemented indirect utility is

$$V_{d,t} = \begin{cases} \frac{1}{\eta} \left(\frac{y_{d,t}^{pc}}{\mathcal{P}_{d,t}} \right)^\eta - \sum_j v_j \log P_{dj,t}, & \eta \neq 0, \\ \log \left(\frac{y_{d,t}^{pc}}{\mathcal{P}_{d,t}} \right) - \sum_j v_j \log P_{dj,t}, & \eta = 0. \end{cases} \quad (3)$$

3.2 Migration

Lagged population $L_{d,t-1}$ enters destination utility through congestion:

$$g_{d,t} = L_{d,t-1}^\iota. \quad (4)$$

The common component of utility from moving to d is

$$U_{d,t}^{com} = \bar{V}_{d,t} g_{d,t} V_{d,t}. \quad (5)$$

An agent starting in origin o values destination d at

$$U_{od,t} = \frac{U_{d,t}^{com}}{\delta_{od,t}}. \quad (6)$$

Migration shares are multinomial-logit in levels:

$$\mu_{od,t} = \frac{U_{od,t}^\nu}{\sum_{d' \in \mathcal{N}} U_{od',t}^\nu}. \quad (7)$$

Population evolves according to

$$L_{d,t} = \sum_{o \in \mathcal{N}} \mu_{od,t} L_{o,t-1}. \quad (8)$$

3.3 Non-homothetic final demand

Expenditure shares in destination d are

$$\psi_{dj,t} = \alpha_j + v_j \left(\frac{y_{d,t}^{pc}}{\mathcal{P}_{d,t}} \right)^{-\eta}. \quad (9)$$

Final consumption expenditure is

$$E_{dj,t}^F = \psi_{dj,t} y_{d,t}^{pc} L_{d,t}. \quad (10)$$

4 Production and firm heterogeneity

4.1 Unit costs and agglomeration

Agglomeration is sector specific:

$$f_{oj,t} = L_{o,t-1}^{\rho_j}. \quad (11)$$

For the baseline technology, unit cost in region o and sector j is

$$c_{oj,t} = \left[\frac{1}{\gamma_{oj,t}} \left(\frac{r_{o,t}}{\beta_{oj,t}} \right)^{\beta_{oj,t}} \left(\frac{w_{o,t}}{1 - \beta_{oj,t}} \right)^{1 - \beta_{oj,t}} \right]^{\gamma_{oj,t}} \prod_{k \in \mathcal{J}} \left(\frac{P_{ok,t}}{\gamma_{ojk,t}^{IO}} \right)^{\gamma_{ojk,t}^{IO}}. \quad (12)$$

The mechanized agricultural technology has an analogous unit cost $\check{c}_{oj,t}$ built from $(\check{\beta}, \check{\gamma}, \check{\gamma}^{IO})$.

4.2 Productivity draws

Potential entrants draw idiosyncratic productivity $\phi \in [1, \kappa]$ from a bounded Pareto distribution with cdf

$$G(\phi) = \frac{1 - \phi^{-\theta}}{1 - \kappa^{-\theta}}, \quad \phi \in [1, \kappa], \quad (13)$$

and density

$$g(\phi) = \frac{\theta \phi^{-\theta-1}}{1 - \kappa^{-\theta}}. \quad (14)$$

The main integral used by the code is

$$I(a, b) = \int_a^b \phi^{\sigma-1} g(\phi) d\phi. \quad (15)$$

4.3 Demand shifter

Given destination expenditures and prices, origin o faces sector- j demand shifter

$$S_{oj,t} = \sum_{d \in \mathcal{N}} \tau_{odj,t}^{1-\sigma} P_{dj,t}^{\sigma-1} E_{dj,t}. \quad (16)$$

4.4 Entry, adoption, and export cutoffs

The zero-profit entry cutoff is implemented as

$$\bar{\phi}_{oj,t} = \max \left\{ 1, \frac{\sigma}{\sigma-1} \frac{1 - s_{oj,t}}{A_{oj,t} f_{oj,t}} \sigma^{\frac{1}{\sigma-1}} c_{oj,t} \left(\frac{F_{oj,t}}{S_{oj,t}} \right)^{\frac{1}{\sigma-1}} \right\}. \quad (17)$$

For agriculture, the adoption cutoff uses the mechanization benefit term

$$B_{oj,t} = \left[\left(\xi \frac{c_{oj,t}}{\check{c}_{oj,t}} \right)^{\sigma-1} - 1 \right]^{\frac{1}{1-\sigma}}, \quad (18)$$

whenever the expression in brackets is positive. Then

$$\check{\phi}_{oj,t} = \max \left\{ 1, \frac{\sigma}{\sigma-1} \frac{1 - s_{oj,t}}{A_{oj,t} f_{oj,t}} \sigma^{\frac{1}{\sigma-1}} c_{oj,t} B_{oj,t} \left(\frac{\check{F}_{oj,t}}{S_{oj,t}} \right)^{\frac{1}{\sigma-1}} \right\}. \quad (19)$$

If the benefit term is non-positive, the code sets $\check{\phi}_{oj,t} = \infty$ and mechanized adoption never occurs.

The export cutoff is

$$\tilde{\phi}_{oj,t} = \max \left\{ 1, \frac{\sigma}{\sigma-1} \frac{1-s_{oj,t}}{A_{oj,t} f_{oj,t}} \sigma^{\frac{1}{\sigma-1}} c_{oj,t} \left(\frac{\tilde{F}_{oj,t}}{\tilde{\tau}_{oj,t}^{1-\sigma} \tilde{D}_{j,t}} \right)^{\frac{1}{\sigma-1}} \times \chi_{oj,t} \right\}, \quad (20)$$

where $\chi_{oj,t} = (\check{c}_{oj,t}/c_{oj,t})/\xi$ for agriculture and $\chi_{oj,t} = 1$ otherwise.

4.5 Productivity aggregates

Define

$$Z_{oj,t} = A_{oj,t} f_{oj,t} I(\bar{\phi}_{oj,t}, \bar{b}_{oj,t})^{\frac{1}{\sigma-1}}, \quad (21)$$

$$\check{Z}_{oj,t} = A_{oj,t} f_{oj,t} I(\check{\phi}_{oj,t}, \kappa)^{\frac{1}{\sigma-1}}, \quad (22)$$

$$\tilde{Z}_{oj,t} = A_{oj,t} f_{oj,t} \chi_j^x I(\tilde{\phi}_{oj,t}, \kappa)^{\frac{1}{\sigma-1}}, \quad (23)$$

where $\bar{b}_{oj,t} = \min\{\check{\phi}_{oj,t}, \kappa\}$ in agriculture and $\bar{b}_{oj,t} = \kappa$ otherwise, and $\chi_j^x = \xi$ for agriculture and 1 otherwise.

5 Sectoral price indices and revenues

5.1 Price indices

Let

$$\Omega_{oj,t} = M_{j,t} \left(\frac{\sigma}{\sigma-1} (1-s_{oj,t}) \right)^{1-\sigma}. \quad (24)$$

The baseline and mechanized domestic supply terms are

$$B_{oj,t}^{trad} = \Omega_{oj,t} \left(\frac{Z_{oj,t}}{c_{oj,t}} \right)^{\sigma-1}, \quad (25)$$

$$B_{oj,t}^{new} = \begin{cases} \Omega_{oj,t} \left(\frac{\check{Z}_{oj,t}}{\check{c}_{oj,t}} \right)^{\sigma-1}, & \text{agriculture,} \\ 0, & \text{otherwise.} \end{cases} \quad (26)$$

Then the destination- d price index in sector j satisfies

$$P_{dj,t}^{1-\sigma} = \sum_{o \in \mathcal{N}} \left(B_{oj,t}^{trad} + B_{oj,t}^{new} \right) \tau_{odj,t}^{1-\sigma} + (\tilde{\tau}_{dj,t} \tilde{p}_{j,t})^{1-\sigma}. \quad (27)$$

5.2 Domestic and foreign revenues

Domestic revenues are

$$R_{oj,t}^{trad} = B_{oj,t}^{trad} S_{oj,t}, \quad (28)$$

$$R_{oj,t}^{new} = B_{oj,t}^{new} S_{oj,t}, \quad (29)$$

$$R_{oj,t}^{dom} = R_{oj,t}^{trad} + R_{oj,t}^{new}. \quad (30)$$

Export revenue is

$$R_{oj,t}^x = M_{j,t} \left(\frac{\sigma}{\sigma-1} (1 - s_{oj,t}) \tilde{\tau}_{oj,t} \right)^{1-\sigma} \left(\frac{\tilde{Z}_{oj,t}}{c_{oj,t}^x} \right)^{\sigma-1} \tilde{D}_{j,t}, \quad (31)$$

where $c_{oj,t}^x = \check{c}_{oj,t}$ for agriculture and $c_{oj,t}^x = c_{oj,t}$ otherwise. Gross output is

$$Y_{oj,t} = R_{oj,t}^{dom} + R_{oj,t}^x. \quad (32)$$

The mass of active firms, exporters, and adopters is

$$N_{oj,t}^{firm} = M_{j,t} [1 - G(\bar{\phi}_{oj,t})], \quad (33)$$

$$N_{oj,t}^x = M_{j,t} [1 - G(\tilde{\phi}_{oj,t})], \quad (34)$$

$$N_{oj,t}^{adopt} = \begin{cases} M_{j,t} [1 - G(\check{\phi}_{oj,t})], & \text{agriculture,} \\ 0, & \text{otherwise.} \end{cases} \quad (35)$$

6 Goods markets and factor markets

6.1 Expenditure system

Under CES demand with constant markups, variable costs are a fixed share of revenues:

$$VC_{oj,t} = \frac{\sigma-1}{\sigma} (R_{oj,t}^{trad} + R_{oj,t}^{new} + R_{oj,t}^x), \quad (36)$$

$$VC_{oj,t}^{trad} = \frac{\sigma-1}{\sigma} R_{oj,t}^{trad}, \quad (37)$$

$$VC_{oj,t}^{new} = \frac{\sigma-1}{\sigma} (R_{oj,t}^{new} + R_{oj,t}^x). \quad (38)$$

Local intermediate demand for sector j in region o is constructed from sectoral variable costs:

$$E_{oj,t}^M = \sum_{k \in \mathcal{J}} VC_{ok,t} \gamma_{okj,t}^{IO}, \quad (39)$$

except in agriculture, where the code replaces the baseline input share on the agricultural output block with

$$VC_{oa,t}^{trad} \gamma_{oak,t}^{IO} + VC_{oa,t}^{new} \gamma_{oak,t}^{IO}. \quad (40)$$

Total expenditure is

$$E_{oj,t} = E_{oj,t}^F + E_{oj,t}^M. \quad (41)$$

6.2 Factor prices

Labor income equals the labor share of variable costs and land income equals the land share of variable costs. For non-agriculture,

$$w_{o,t} L_{o,t} = \sum_j (1 - \beta_{oj,t}) \gamma_{oj,t} VC_{oj,t}, \quad (42)$$

$$r_{o,t} H_{o,t} = \sum_j \beta_{oj,t} \gamma_{oj,t} VC_{oj,t}. \quad (43)$$

For agriculture, the code splits baseline and mechanized production:

$$w_{o,t}L_{o,t} = (1 - \beta_{oa,t})\gamma_{oa,t}VC_{oa,t}^{trad} + (1 - \check{\beta}_{oa,t})\check{\gamma}_{oa,t}VC_{oa,t}^{new} + \sum_{j \neq a} (1 - \beta_{oj,t})\gamma_{oj,t}VC_{oj,t}, \quad (44)$$

$$r_{o,t}H_{o,t} = \beta_{oa,t}\gamma_{oa,t}VC_{oa,t}^{trad} + \check{\beta}_{oa,t}\check{\gamma}_{oa,t}VC_{oa,t}^{new} + \sum_{j \neq a} \beta_{oj,t}\gamma_{oj,t}VC_{oj,t}. \quad (45)$$

7 Government budget and profit rebate

Aggregate wage income is

$$W_t = \sum_{o \in \mathcal{N}} w_{o,t}L_{o,t}. \quad (46)$$

Aggregate profits are

$$\Pi_t = \sum_{o \in \mathcal{N}} \sum_{j \in \mathcal{J}} \left[\frac{1}{\sigma} Y_{oj,t} - N_{oj,t}^{firm} F_{oj,t} - N_{oj,t}^x \tilde{F}_{oj,t} - N_{oj,t}^{adopt} \check{F}_{oj,t} \right]. \quad (47)$$

The profit rebate is

$$\bar{\pi}_t = \frac{\Pi_t}{W_t}. \quad (48)$$

Subsidy and infrastructure spending imply

$$\bar{\tau}_t = \frac{\sum_{o,j} \frac{\sigma-1}{\sigma} s_{oj,t} Y_{oj,t} + G_t^{infra}}{W_t}, \quad (49)$$

where G_t^{infra} is exogenous road and building expenditure.

8 Static equilibrium

Given L_{t-1} and exogenous period- t fundamentals, a static equilibrium is a tuple

$$\{w_{o,t}, r_{o,t}, P_{oj,t}, E_{oj,t}, L_{o,t}, \bar{\tau}_t, \bar{\pi}_t\}_{o \in \mathcal{N}, j \in \mathcal{J}}$$

such that:

1. household migration and final demand satisfy the equations in Section 2,
2. firm cutoffs, price indices, and revenues satisfy the equations in Sections 3 and 4,
3. expenditure balances satisfy $E_{oj,t} = E_{oj,t}^F + E_{oj,t}^M$,
4. factor markets satisfy the wage and land-rent equations in Section 5,
5. $\bar{\pi}_t$ and $\bar{\tau}_t$ satisfy the aggregate budget formulas in Section 6.

The code treats the static equilibrium as a fixed point in

$$x_t = (w_t, r_t, P_t, E_t, \bar{\tau}_t, \bar{\pi}_t),$$

with L_t implied from x_t and L_{t-1} .

9 Dynamic path

The full dynamic equilibrium is computed sequentially because the only endogenous state variable is lagged population:

$$L_{t-1} \longrightarrow \text{solve static equilibrium at } t \longrightarrow L_t. \quad (50)$$

For $t = 1$, the state is the initial population vector L_0 . For $t \geq 2$, the state is the solved population vector from period $t - 1$.

10 Numerical solution

The solver uses a damped fixed-point iteration:

1. start from a positive guess for $(w, r, P, E, \bar{\tau}, \bar{\pi})$,
2. compute the implied mapping $F(x_t)$ from the equilibrium equations,
3. update each component with geometric damping in logs,
4. stop when the maximum absolute log residual falls below tolerance.

For example, wages are updated by

$$w^{new} = \exp \left[(1 - \lambda) \log w^{old} + \lambda \log w^{imp} \right], \quad (51)$$

and analogously for r , P , E , $\bar{\tau}$, and $\bar{\pi}$. Later periods are warm-started from the previous period's solution.

11 Implementation restrictions

The code enforces several restrictions that are useful to keep in mind when extending the model:

- L_0 is normalized to sum to one.
- $\gamma_{oj,t} + \sum_k \gamma_{ojk,t}^{IO} = 1$ for every (o, j, t) .
- $\check{\gamma}_{oj,t} + \sum_k \check{\gamma}_{ojk,t}^{IO} = 1$ for every (o, j, t) .
- $\check{\gamma}_{oj,t} \check{\beta}_{oj,t} = \gamma_{oj,t} \beta_{oj,t}$, so the land share inside value added is preserved across agricultural technologies.
- Outside the heavy-manufacturing input, the mechanized agricultural input shares are restricted to equal the baseline input shares.
- Subsidy rates satisfy $0 \leq s_{oj,t} < 1$.

12 Interpretation

The model combines five ingredients:

1. non-homothetic demand and endogenous migration,
2. Melitz-style firm heterogeneity with bounded Pareto draws,
3. input-output linkages across sectors,
4. sector-specific agglomeration from lagged population,
5. policy wedges acting through productivity, fixed costs, subsidies, trade costs, migration costs, and foreign demand.

That is the economic system solved by the repository. Any empirical exercise or policy counterfactual run through the package is a perturbation of this equilibrium system.